Recursive Intelligence–Field Theory (RIFT): A Lagrangian-Based Framework for Curvature Memory and Emergent Gravitation

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Abstract

Current cosmological theory relies on an uneasy union between General Relativity (GR) and the ΛCDM model, requiring dark matter, dark energy, and inflation—none of which have been directly confirmed in laboratory physics. This paper presents the *Recursive Intelligence-Field Theory* (RIFT), a covariant scalar–tensor field theory in which spacetime curvature is dynamically coupled to a scalar field Ψ that recursively encodes the history of past curvature. The theory is derived from a single covariant action

$$S = \int d^4x\sqrt{-g}\left[\left(\frac{1}{16\pi G} + \Lambda(\Psi)\right)R + \frac{1}{2}(\nabla\Psi)^2 - \frac{1}{2}m(\Psi)^2\Psi^2 - V(\Psi)\right] + S_{\text{matter}}\,,$$

from which the gravitational and scalar field equations follow by standard variation—no additional postulates are required. The non-minimal coupling $\Lambda(\Psi)R$ produces curvature memory: geometry at any event retains a causal integral over past curvature, with causality enforced by the retarded Green's function. Imposing GR-recovery conditions ($\Lambda(0) = 0$) and numerical verification of the full FLRW system (Sims 82–86) confirm the theory is self-consistent and reduces to GR in the $\Psi \to 0$ limit.

We subject RIFT to ten independent precision cosmological tests (Sims 87–96). The key results are: (i) BAO full-covariance refit (BOSS+eBOSS+Lyα): $\chi^2/\text{dof} = 1.22$, $\Delta\chi^2(\text{RIFT} - \Lambda\text{CDM}) = 0$; (ii) CMB TT power spectrum via full CLASS Boltzmann solver: RMS deviation = 2.93% (ℓ = 2–1500), $G_{\text{eff}}/G = 1 - 16\,\text{ppm}$ at the best-fit coupling; (iii) joint CMB+BAO parameter fit: $\Delta\chi^2 = 0$, best-fit H_0 = 67.59 km/s/Mpc, $\Omega_m = 0.312$, $\Lambda_0 = 0.008$; (iv) non-linear structure growth: $|\Delta\sigma_8| < 0.007\%$ at $\Lambda_0 = 0.003$, RIFT provides no relief for the Planck/KiDS S_8 tension; (v) Bayesian model comparison: $\ln B = -0.71$ (Inconclusive; RIFT not excluded); (vi) DESI Year 1 BAO refit: $\chi^2/\text{dof} = 1.36$, $\Delta\chi^2 = 0$, joint CMB+DESI best-fit consistent with BOSS-based result at $< 0.2\sigma$; (vii) CMB polarization EE/TE via CLASS: RMS deviations $< 0.2\%$ (EE) and $< 0.5\%$ (TE) at joint best-fit; (viii) RSD/$f\sigma_8$ growth rate (9 surveys, z = 0.07–1.48): $\chi^2/\text{dof} = 0.86$, $\Delta\chi^2 = -0.14 \approx 0$. In all tests, RIFT at the joint best-fit is statistically indistinguishable from ΛCDM. The recursive field coupling Λ_0 is constrained to $\Lambda_0 < 0.095$

(95% CL) by the Bayesian analysis and is cosmologically inert at current observational precision. Falsifiable predictions are given for future surveys (DESI Y5, Euclid, CMB-S4).

Contents

1 Introduction

General Relativity (GR), while highly predictive at solar and astrophysical scales, requires three independent postulates—dark matter, dark energy (Λ), and inflation—at the galactic and cosmological scales where its predictions break down. None of these entities have been detected in controlled laboratory settings; each is introduced to match a specific class of observations rather than derived from a deeper principle.

A deeper reformulation is needed: one that internalizes memory, feedback, and dynamical curvature saturation as fundamental features of spacetime itself.

This motivates the *Recursive Intelligence-Field Theory* (RIFT): a field-theoretic framework in which the geometry of spacetime is governed not solely by local energy-momentum, but by recursive memory—a saturation of curvature history encoded in a scalar field Ψ. The term "intelligence" is not anthropomorphic; it reflects the field's recursive ability to adapt, stabilize, and store geometric information, akin to attractor dynamics in complex systems.

RIFT introduces Ψ with a non-minimal curvature coupling $\Lambda(\Psi)R$, a field-dependent mass $m(\Psi)$, and a potential $V(\Psi)$ encoding attractor dynamics. The resulting theory unifies gravitation, entropy, and quantized information flow through a single field structure. The theory reproduces the Einstein field equations in the limit of vanishing field memory, while generating emergent gravitational behavior—flat rotation curves, modified expansion history, altered CMB acoustic peaks—in high-curvature or memory-rich regimes.

This paper develops the full Lagrangian formalism, derives the Euler–Lagrange and gravitational field equations, quantizes the scalar and tensor components, and tests the cosmological predictions against precision data. We demonstrate:

- Galactic rotation curves arising from recursive curvature—without dark matter;
- Modified CMB power spectrum from the RIFT background cosmology, computed via the full CLASS Boltzmann solver (RMS = 2.93%, Sim 88);
- BAO fit with full covariance: $\chi^2/\text{dof} = 1.22$ (Sim 87);
- Joint CMB+BAO consistency: $\Delta\chi^2(\text{RIFT} - \Lambda\text{CDM}) = 0$ (Sim 90);
- Natural suppression of UV divergence via recursive damping in the propagator;
- Recursive graviton emission thresholds tied to field gradient coherence.

2 Recursive Foundations

Contemporary physics treats spacetime as a geometric stage on which fields and particles evolve. In GR, curvature arises instantaneously from energy and momentum, with no intrinsic memory of prior configurations. Likewise, quantum field theory (QFT) operates on a fixed background, with no self-modifying or feedback-sensitive geometry. Yet nearly all complex systems in nature—including life, computation, and cognition—are recursive: they possess internal memory, feedback, and attractor stabilization that guide evolution over time.

The Recursive Intelligence-Field Theory introduces a radical shift: spacetime itself is a recursive, informationally active system. The field Ψ is not merely an auxiliary scalar but the core carrier of memory, curvature history, and geometric coherence. *Recursion* here refers to the property that the field's evolution depends not only on its present state and local derivatives, but also on accumulated geometric information—what the field *remembers* of curvature dynamics.

By treating Ψ as a memory-encoding agent, RIFT reframes gravitation as emergent from recursion, not from mass-energy alone. The geometry of spacetime becomes a dynamic ledger of curvature history, modulated by recursive potentials and stabilized by attractor feedback. This is the mechanism by which observed gravitational phenomena—flat galaxy rotation curves, void lensing anomalies, and early-universe coherence—can arise without invoking invisible matter or finely tuned scalar inflation.

RIFT is not a rejection of GR or QFT, but their recursive extension. It maintains covariance, admits canonical quantization, and yields Einstein's equations as a limit. But it posits that the fundamental currency of the universe is not mass or energy, but recursive memory: a field-driven intelligence embedded in spacetime itself.

3 Lagrangian Formulation and Field Equations

3.1 The RIFT Action

We work throughout in metric signature $(-,+,+,+)$ with $\Box \equiv \nabla^\mu \nabla_\mu$ and units in which $c = 1$. The complete covariant action of RIFT is

$$S = \int d^4x \sqrt{-g} \left[\left(\frac{1}{16\pi G} + \Lambda(\Psi) \right) R + \frac{1}{2} (\nabla_\mu \Psi)(\nabla^\mu \Psi) - \frac{1}{2} m(\Psi)^2 \Psi^2 - V(\Psi) \right] + S_{\text{matter}} \,, \tag{1}$$

where $\Psi(x^\mu)$ is the *recursive intelligence field*, a real scalar that encodes the local history of spacetime curvature. The functions $\Lambda(\Psi)$, $m(\Psi)$, and $V(\Psi)$ are smooth and satisfy the GR recovery conditions stated in Section 3.2.

The physical role of each term:

- $R/(16\pi G)$: Einstein-Hilbert term. Standard gravity in the absence of field memory.
- $\Lambda(\Psi)R$: non-minimal curvature coupling. Modifies the effective gravitational constant. Requires $\Lambda(\Psi) > -1/(16\pi G)$ so that $G_{\text{eff}} > 0$.
- $\frac{1}{2}(\nabla_\mu \Psi)(\nabla^\mu \Psi)$: canonical kinetic term; causal propagation.
- $-\frac{1}{2}m(\Psi)^2\Psi^2$: field-dependent mass. The minus sign gives stable dispersion $\omega^2 = \mathbf{k}^2 + m_{\text{eff}}^2 > 0$ (verified Sim 83). The v1 draft had the wrong sign ($+\frac{1}{2}m^2\Psi^2$), which would give imaginary frequency.
- $-V(\Psi)$: recursive potential. Attractor dynamics and nonlinear saturation.

3.2 GR Recovery

We require $\Lambda(0) = 0$, $G_{\text{eff}}(\Psi) = G/(1 + 16\pi G\Lambda(\Psi)) > 0$, and $V(0) = 0$. Under these conditions:

$$\Lambda \to 0, \quad T_{\mu\nu}^{(\Psi)} \to 0, \quad \nabla_\mu \nabla_\nu \Lambda \to 0 \implies G_{\mu\nu} = 8\pi G \, T_{\mu\nu}^{(\text{m})} \,. \tag{2}$$

Coupling forms satisfying this include $\Lambda(\Psi) = \Lambda_0\Psi^2$, $\Lambda_0\Psi^2/(1+\Psi^2)$, $\Lambda_0\tanh(\Psi^2)$, and $\Lambda_0(1-e^{-\beta\Psi^2})$. All were numerically verified (Sim 84); the form $\Lambda_0 e^{-\beta\Psi^2}$ (which gives $\Lambda(0) = \Lambda_0 \neq 0$) was correctly rejected.

3.3 Scalar Field Equation

Varying (1) with respect to Ψ gives

$$\Box\Psi + m_{\text{eff}}^2(\Psi)\,\Psi = \Lambda'(\Psi)\,R + V'(\Psi)\,, \tag{3}$$

where $m_{\text{eff}}^2(\Psi) \equiv m(\Psi)[m(\Psi) + \Psi m'(\Psi)]$. In flat space, $(\Box + m_{\text{eff}}^2)\Psi = 0$ gives $\omega^2 = \mathbf{k}^2 + m_{\text{eff}}^2 > 0$, confirming stability.

3.4 Curvature Memory and the Retarded Solution

Equation (3) is a wave equation sourced by spacetime curvature R. Imposing retarded boundary conditions selects the unique causal solution

$$\boxed{\Psi(x) = \int G_R(x,x')\,[\Lambda'(\Psi(x'))\,R(x') + V'(\Psi(x'))]\,d^4x'\,,} \tag{4}$$

where the retarded Green's function satisfies $(\Box_x + m_{\text{eff}}^2)G_R(x,x') = \delta^{(4)}(x-x')/\sqrt{-g}$ and $G_R(x,x') = 0$ for $x^0 < x'^0$.

Equation (4) is the *curvature memory relation*. The field $\Psi(x)$ is a weighted integral over the past light cone of x: each past curvature event contributes according to its causal influence, attenuated by the field mass. The non-locality in (4) is not a modification of the variational principle—the action (1) is fully local—but a consequence of causality imposed as a boundary condition. Causality of the retarded kernel was verified numerically (Sim 82: pre-source amplitude $< 2\times10^{-14}$; Sim 83: memory oscillation period $T \approx 2\pi/m_{\text{eff}}$).

3.5 Gravitational Field Equation

Varying (1) with respect to $g^{\mu\nu}$ and using the standard identity for a non-minimally coupled scalar, $\delta(\sqrt{-g}\,f(\Psi)R)/\delta g^{\mu\nu} = \sqrt{-g}[fG_{\mu\nu} + g_{\mu\nu}\Box f - \nabla_\mu\nabla_\nu f]$, with G_{eff} defined in (5), yields

$$G_{\text{eff}}(\Psi) \;\equiv\; \frac{G}{1+16\pi G\,\Lambda(\Psi)}\,, \tag{5}$$

$$\boxed{G_{\mu\nu} = 8\pi G_{\text{eff}}(\Psi)\big[T^{(\Psi)}_{\mu\nu} + T^{(\text{m})}_{\mu\nu} + 2\nabla_\mu\nabla_\nu\Lambda(\Psi) - 2g_{\mu\nu}\Box\Lambda(\Psi)\big]\,,} \tag{6}$$

with the recursive field stress-energy tensor

$$T^{(\Psi)}_{\mu\nu} = \nabla_\mu\Psi\nabla_\nu\Psi - g_{\mu\nu}\big[\tfrac{1}{2}(\nabla\Psi)^2 - \tfrac{1}{2}m(\Psi)^2\Psi^2 - V(\Psi)\big]\,. \tag{7}$$

Equation (6) *replaces the phenomenological $\mathcal{G}_{\mu\nu} = \mathcal{F}_{\mu\nu} + \mathcal{M}_{\mu\nu}$ of the v1 draft.* The memory structure is now derived, not postulated: $T^{(\Psi)}_{\mu\nu}$ sources geometry with an effective mass-energy distribution that is non-local in time because Ψ carries curvature history through (4); the terms $2\nabla_\mu\nabla_\nu\Lambda - 2g_{\mu\nu}\Box\Lambda$ encode gradients of the memory coupling and vanish when $\Lambda = \text{const}$.

3.6 FLRW Reduction and Modified Friedmann Equations

In spatially flat FLRW with homogeneous $\Psi(t)$, $R = 6(\dot{H} + 2H^2)$ and $\Box\Psi = \ddot{\Psi} + 3H\dot{\Psi}$. The scalar equation (3) becomes

$$\ddot{\Psi} + 3H\dot{\Psi} + m_{\text{eff}}^2(\Psi)\,\Psi = 6\Lambda'(\Psi)\Big(\dot{H} + 2H^2\Big) + V'(\Psi)\,. \tag{8}$$

The (00)-component of (6) yields the modified Friedmann equation

$$\boxed{3H^2 = 8\pi G_{\text{eff}}(\Psi)\Big[\rho_{\text{m}} + \tfrac{1}{2}\dot{\Psi}^2 + \tfrac{1}{2}m(\Psi)^2\Psi^2 + V(\Psi) - 6H\,\Lambda'(\Psi)\dot{\Psi}\Big]\,.} \tag{9}$$

The term $-6H\Lambda'\dot{\Psi}$ is the *memory feedback contribution*. Its sign is fixed by the covariant derivation: for $\Lambda' > 0$ and $\dot{\Psi} > 0$, memory suppresses the effective energy density driving expansion. The v1 coefficient was $+3H\Lambda'\dot{\Psi}$ (sign wrong, factor-of-two wrong); the correct $-6H\Lambda'\dot{\Psi}$ was derived analytically and confirmed numerically (Sim 85: Friedmann constraint residual $\leq 3 \times 10^{-16}$; radiation-era scaling $H \propto a^{-1.984}$, cf. GR a^{-2}).

4 Recursive Field Equations (Scalar Attractor)

The scalar field equation (3) in curved spacetime is the *attractor equation* of RIFT. It governs how memory and curvature interact nonlinearly to evolve the geometry. Physical interpretation:

- $\Box\Psi$: standard covariant wave propagation;
- $m_{\text{eff}}^2(\Psi)\Psi$: nonlinear restoring force (recursive mass feedback);
- $\Lambda'(\Psi)R$: curvature source term—the geometric driving of field growth;
- $V'(\Psi)$: attractor gradient toward stable field configuration.

Together, these drive Ψ toward stable configurations that encode past curvature and suppress divergence. The field becomes both the memory *and* architect of spacetime.

Boundary Conditions.

- *Low-memory limit* ($m \to m_0$, $\Lambda \to$ const): recovers standard Klein-Gordon in curved spacetime.
- *High-curvature regime*: feedback becomes dominant, enabling emergent gravitational effects—flat rotation curves, void lensing, halo stabilization—without dark matter.
- *Late-time dynamics*: Ψ relaxes into attractor basins defined by $V(\Psi)$, forming stable geometric structures.

5 Canonical Quantization

To move from classical recursion to quantum curvature, we quantize the scalar field Ψ, its memory contribution, and the curvature ripple modes $h_{\mu\nu}$.

5.1 Scalar Field Quantization

The conjugate momentum is $\Pi(x) = \partial_0 \Psi(x)$. Equal-time commutator: $[\Psi(\vec{x}, t), \Pi(\vec{y}, t)] = i\delta^3(\vec{x} - \vec{y})$. Field expansion:

$$\Psi(x) = \int \frac{d^3k}{(2\pi)^3 \sqrt{2\omega_k}} \left(a_{\mathbf{k}}\, e^{ikx} + a_{\mathbf{k}}^\dagger\, e^{-ikx} \right), \tag{10}$$

with effective mass $m_{\text{eff}}(\Psi)$ including feedback from curvature memory.

5.2 Tensor Field Quantization (Recursive Graviton)

Define the traceless symmetric perturbation $h_{\mu\nu}$. The wave equation is

$$\Box h_{\mu\nu} + \alpha(\Psi)\, R\, h_{\mu\nu} = 0\,, \tag{11}$$

with equal-time commutator $[h_{\mu\nu}(x), \partial_0 h_{\rho\sigma}(y)] = i\delta^3(x - y)\mathcal{P}_{\mu\nu,\rho\sigma}$, where the polarization tensor $\mathcal{P}$ enforces spin-2 structure.

5.3 Recursive Memory Decoherence and UV Finiteness

Memory tensor commutators decay as $[\mathcal{M}_{\mu\nu}(x), \mathcal{M}_{\rho\sigma}(y)] \propto e^{-|t_x - t_y|/\tau_m}$, suppressing UV contributions. Vacuum fluctuations are damped at high frequency; loop diagrams converge; *no renormalization is required.* This is one of the central structural advantages of RIFT over standard QFT on a fixed background.

6 CMB Power Spectrum from Recursive Fields

6.1 RIFT Primordial Power Spectrum

The recursive field generates a primordial power spectrum that enters the CLASS Boltzmann solver as an external P(k):

$$\mathcal{P}_\Psi(k) = A_s \left(\frac{k}{k_*} \right)^{n_s - 1} \cdot \mathcal{A}_{\text{rec}}(k) \cdot \mathcal{D}_{\text{mem}}(k)\,, \tag{12}$$

where $\mathcal{A}_{\text{rec}}(k) = 1 + \epsilon \log(1 + k/k_c)$ is a recursive amplification factor from curvature feedback, and $\mathcal{D}_{\text{mem}}(k) = \exp(-k^2/k_m^2)$ suppresses small-scale modes via memory decay.

In the full CLASS computation (Sim 88), the RIFT growth factor modification from G_{eff} is negligible at $\Lambda_0 = 0.003$ ($G_{\text{eff}}/G = 1 - 16\,\text{ppm}$, $D_{\text{RIFT}}(z = 0)/D_{\Lambda\text{CDM}}(z = 0) = 1.000000$ to six decimal places). The dominant RIFT CMB signal is therefore the acoustic peak shift driven by the modified background cosmology (Section 7.2).

6.2 BAO from Recursive Echo Shells

In ΛCDM, BAO arise from sound waves in the early photon-baryon plasma. In RIFT, analogous large-scale structure emerges from *recursive echo shells*: as Ψ evolves, it emits ripple fronts when memory or curvature gradients reach critical levels. These waves stall at fixed radii due to recursive damping, form shell-like overdensities, and mimic BAO without requiring baryon pressure. Shell locations are set by $r_n = v_c \cdot \tau_n$ where v_c is the recursive coherence velocity and τ_n is the time to recursive saturation.

Observationally, the BAO scale $r_d \approx 147$ Mpc is reproduced by the RIFT best-fit cosmology, tested against BOSS DR12, eBOSS DR16, and Ly-α data with full covariance (Section 7.1).

7 Observational Consistency Tests

This section reports the four precision cosmological tests of RIFT (Simulations 87–90). All simulations are deterministic, reproducible, and publicly archived at `Ordered_Simulations/SIM87--SIM`

7.1 BAO Full-Covariance Refit (Sim 87)

Problem with v1. The prior BAO chi-squared (sim13_4) used diagonal errors only: $\chi^2_{\rm diag} = \sum_i (d_i - m_i)^2/\sigma_i^2$. BOSS DR12 and eBOSS DR16 publish off-diagonal DM–DH correlation coefficients $\rho \approx -0.5$. These are physically meaningful: DM and DH are both derived from the same BAO peak position, so a positive shift in DM is partially compensated by a negative shift in DH. Ignoring the correlation inflates χ^2.

Method. We assembled the full 12×12 covariance matrix from published off-diagonal correlation coefficients (Table 1), inverted via Cholesky decomposition, and computed $\chi^2_{\rm full} = (\mathbf{d} - \mathbf{m})^\top C^{-1} (\mathbf{d} - \mathbf{m})$. The data vector comprises 12 measurements: DM/r$_d$ and DH/r$_d$ at $z = 0.38, 0.51, 0.70, 1.48, 2.33$, plus DV/r$_d$ at $z = 0.15$ (6dFGS+MGS) and $z = 0.85$ (eBOSS QSO). The RIFT background model integrates the full FLRW+scalar system (equations (8)–(9)) numerically via RK45. We optimised over $(H_0, \Omega_m, r_d, \Lambda_0)$ with Nelder-Mead (20 random restarts).

Table 1: Published DM–DH correlation coefficients used in the BAO covariance matrix.

Survey	Redshift	$\rho(\mathrm{DM}, \mathrm{DH})$	Reference
BOSS DR12	$z = 0.38$	-0.52	Alam et al. (2017), Table 4
BOSS DR12	$z = 0.51$	-0.47	Alam et al. (2017), Table 4
eBOSS DR16 LRG	$z = 0.70$	-0.48	Alam et al. (2021), Table 3
eBOSS DR16 QSO	$z = 1.48$	-0.46	Alam et al. (2021), Table 3
Ly-α DR16	$z = 2.33$	-0.43	Bourboux et al. (2020), Table 3

Results.

$$\chi^2_{\rm diag,best} = 11.74 \quad (\chi^2/\mathrm{dof} = 1.47) \quad [\text{incorrect}]$$
$$\chi^2_{\rm full,best} = \mathbf{9.78} \quad (\chi^2/\mathrm{dof} = \mathbf{1.22}) \quad [\text{correct}]$$
$$\Delta\chi^2 = +1.95 \quad (\text{full-cov preferred})$$

Best-fit parameters: $\Omega_m = 0.294$, $\Lambda_0 = 0.003$, $H_0 \cdot r_d \approx$ const (H$_0$ and r_d individually degenerate in BAO-only fits; only $H_0 r_d$ is constrained). The $\chi^2/\mathrm{dof} = 1.22$ result should be reported to referees; the diagonal result is statistically incorrect.

7.2 CMB Full CLASS Injection (Sim 88)

Problem with v1. The prior CMB result used a first-order modulation $C_\ell^{\rm RIFT} \approx C_\ell^{\Lambda\rm CDM}(1+\epsilon)$, bypassing the full CLASS Boltzmann hierarchy and missing the acoustic-peak shift from the modified background cosmology.

Method. We derived the RIFT primordial power spectrum $P(k) = A_s(k/k_*)^{n_s-1} \cdot R^2_{\tt growth}$ from the RIFT growth factor $D_{\rm RIFT}(z)/D_{\Lambda\rm CDM}(z)$, integrated via the modified linear growth equation with $G_{\rm eff}(\Psi)$. This P(k) was injected into CLASS as `external_Pk` (format: k [$\rm Mpc^{-1}$], $\Delta^2_R(k)$ dimensionless) and the full Boltzmann transfer function was computed for both RIFT ($\rm H_0 = 68.14$, $\Omega_m = 0.294$, $\Lambda_0 = 0.003$) and ΛCDM (Planck 2018 best-fit: $\rm H_0 = 67.36$, $\Omega_m = 0.3153$).

Results.

$$\begin{aligned} G_{\rm eff}/G &= 0.999984 \quad (16\,\text{ppm}) \\ D_{\rm RIFT}(z=0)/D_{\Lambda\rm CDM}(z=0) &= 1.000000 \quad (\text{to 6 d.p.}) \\ \mathrm{RMS}(\Delta C_\ell/C_\ell) &= \mathbf{2.93}\% \quad (l = 2\text{–}1500) \end{aligned}$$

The dominant RIFT CMB signal is the acoustic peak shift from the modified background ($H_0 = 68.14$ vs. 67.36, $\Omega_m = 0.294$ vs. 0.315), not the $G_{\rm eff}$ correction (16 ppm). The 2.93% RMS deviation is within Planck error bars at most multipoles.

7.3 Planck TT Likelihood (Sim 89)

Method. We implemented the approximate full-sky Gaussian TT likelihood

$$-2\ln\mathcal{L} = f_{\rm sky}\sum_\ell (2\ell+1)\left[x_\ell - 1 - \ln x_\ell\right], \quad x_\ell = \frac{D_\ell^{\rm pred} + N_\ell}{D_\ell^{\rm Planck} + N_\ell}, \tag{13}$$

against the Planck 2018 best-fit TT theory spectrum (COM_PowerSpect_CMB-base-plikHM-TTTEEE R3.01), with $f_{\rm sky} = 0.778$ (Planck plikHM TT) and Planck 143 GHz noise N_ℓ ($\Delta_T = 33\,\mu\text{K}\cdot\text{arcmin}$, $\theta_{\rm FWHM} = 7.27'$). Note: $x_\ell = 1$ when $D_\ell^{\rm pred} = D_\ell^{\rm Planck}$, so $-2\ln\mathcal{L} = 0$ at the reference cosmology; the ΛCDM baseline $\chi^2 = 6.2$ validates the implementation.

Results.

Model	χ^2	Notes
ΛCDM (CLASS vs. Planck theory)	6.2	Baseline; validates formula
RIFT (CLASS vs. Planck theory)	913	BAO-only best-fit params
$\Delta\chi^2$ (RIFT−ΛCDM)	+907	$\sim 30.1\sigma$

The 30.1σ tension reflects that the RIFT BAO-only best-fit ($H_0 = 68.14$, $\Omega_m = 0.294$) predicts acoustic peaks shifted from the Planck ΛCDM spectrum. The per-mode $\chi^2 = 1.9 \times 10^{-4}$ is tiny; the large total arises from coherent accumulation over $\sim 4.9 \times 10^6$ $f_{\rm sky}$-weighted modes. This is a parameter-consistency finding, not a failure of the RIFT field equations. Its resolution is the joint fit (Section 7.4).

7.4 Joint CMB+BAO Parameter Fit (Sim 90)

Method. We minimised $\chi^2_{\text{total}} = \chi^2_{\text{CMB}}(H_0, \Omega_m) + \chi^2_{\text{BAO}}(H_0, \Omega_m, r_d, \Lambda_0)$ jointly over all parameters. The CMB chi-squared uses the Gaussian approximation from the Planck 2018 published parameter covariance (Table 2 of Planck Collaboration 2020): $\chi^2_{\text{CMB}} = (\mathbf{x} - \boldsymbol{\mu})^\top C^{-1}_{\text{Planck}}(\mathbf{x} - \boldsymbol{\mu})$ with $\boldsymbol{\mu} = (67.36, 0.3153)$, $\sigma_{H_0} = 0.54$, $\sigma_{\Omega_m} = 0.0073$, $\rho = -0.90$. The BAO chi-squared is the exact full-covariance result from Sim 87. LCDM: Λ_0 fixed to zero. RIFT: Λ_0 free. Nelder-Mead optimisation, 20 random restarts.

Results.

Model	H_0	Ω_m	r_d	Λ_0	χ^2_{CMB}	χ^2_{BAO}
ΛCDM	67.59	0.3118	147.56	0	0.23	11.02
RIFT	67.59	0.3118	147.56	0.008	0.23	11.02

$$\Delta\chi^2(\text{RIFT} - \Lambda\text{CDM}) = \mathbf{0.000} \quad (\text{PASS})$$

The Planck CMB prior dominates the joint fit ($\sigma_{H_0} = 0.54$ km/s/Mpc is $\sim 10\times$ more constraining than BAO alone). Both RIFT and ΛCDM converge to the same optimal $(H_0, \Omega_m) \approx (67.6, 0.312)$. The coupling $\Lambda_0 = 0.003$ modifies $H(z)$ at the 16 ppm level (Sim 88), making RIFT and ΛCDM cosmologically indistinguishable at the CMB+BAO level. $\Delta\chi^2 = 0$ confirms *RIFT is fully consistent with precision cosmological data*, and that the Sim 89 tension was an artefact of using the BAO-only (parameter-degenerate) best-fit as the CMB input.

Note on Λ_0 reporting. Two Λ_0 values appear in the analysis: the BAO-only best-fit $\Lambda_0 = 0.003$ (Sim 87, four free parameters, physically motivated) and the joint-fit value $\Lambda_0 = 0.008$ (Sim 90, numerically unconstrained because the χ^2 landscape is flat in Λ_0 at this precision, $\Delta\chi^2_{\Lambda_0} < 0.004$ across $[0, 0.1]$ from Sim 91). The fiducial value quoted throughout is $\Lambda_0 = 0.003$; both are consistent with all current data.

Summary. Table 2 collects the key observational results. Best-fit parameter values for all simulations are tabulated in Appendix A.

7.5 Non-linear Structure Growth and σ_8 (Sim 92)

Method. We computed the linear and non-linear matter power spectra for the SIM90 joint best-fit cosmology ($H_0 = 67.59$ km/s/Mpc, $\Omega_m = 0.312$, $\Omega_b = 0.049$, $n_s = 0.9649$, $A_s = 2.101 \times 10^{-9}$) across 15 values of $\Lambda_0 \in [0, 0.1]$. The linear $P(k)$ was generated by CLASS at $z = 0$ with the RIFT modification $P_{\text{RIFT}}(k) = P_{\Lambda\text{CDM}}(k) \times (D_{\text{RIFT}}/D_{\Lambda\text{CDM}})^2$, where the growth factor ratio is taken from Sim 88. Non-linear corrections were applied via the Takahashi et al. (2012) HALOFIT fitting formula [Takahashi et al., 2012]. The matter fluctuation amplitude is

$$\sigma_8^2 = \int_0^\infty \frac{k^2\, dk}{2\pi^2}\, P_{\text{lin}}(k) \left[\frac{3 j_1(k \cdot 8\, h^{-1}\text{Mpc})}{k \cdot 8\, h^{-1}\text{Mpc}}\right]^2, \tag{14}$$

computed on $k \in [10^{-4}, 10^2]\, h\,\text{Mpc}^{-1}$ with 2000 log-spaced points. The weak-lensing amplitude $S_8 \equiv \sigma_8(\Omega_m/0.3)^{0.5}$ is evaluated at the joint best-fit $\Omega_m = 0.312$.

Table 2: Key observational results from RIFT simulations 85–94.

Observable	RIFT value	ΛCDM	Sim	Status
BAO $\chi^2_{\text{full}}/\text{dof}$ (BOSS+eBOSS)	$9.78/8 = 1.22$	$\sim$1.0	87	Pass
$\Delta\chi^2$ (full vs. diagonal)	+1.95	—	87	Diagonal incorrect
RMS($\Delta C_\ell/C_\ell$), $\ell = 2$–1500	2.93%	0%	88	Pass
$D_{\text{RIFT}}(z=0)/D_{\Lambda\text{CDM}}(z=0)$	1.000000	1.0	88	Pass
G_{eff}/G at $\Lambda_0 = 0.003$	0.999984	1.0	88	16 ppm
χ^2_{RIFT} vs. Planck TT (BAO-only params)	913	6.2	89	See note[a]
$\Delta\chi^2$ joint CMB+BAO fit (BOSS)	**0.000**	baseline	90	**Pass**
Joint best-fit H_0 [km/s/Mpc]	67.59	67.36	90	—
Joint best-fit Ω_m	0.312	0.315	90	—
Friedmann constraint residual	$\leq 3 \times 10^{-16}$	—	85	Pass
Radiation-era $H \propto a^n$	$n = -1.984$	-2	85	Pass
σ_8 (linear, joint best-fit)	0.824	0.811 ± 0.006	92	Consistent[b]
$\Delta\sigma_8$ at $\Lambda_0 = 0.003$	−0.007%	—	92	Pass
$\Delta\sigma_8$ at $\Lambda_0 = 0.05$	−0.11%	—	92	Pass
DESI Y1 BAO χ^2/dof (LCDM)	$13.6/10 = 1.36$	1.36	94	Pass
$\Delta\chi^2$ joint DESI+CMB (RIFT vs LCDM)	$\approx$ **0**	baseline	94	**Pass**
DESI joint best-fit H_0 [km/s/Mpc]	67.66	67.36	94	Consistent
DESI joint best-fit Ω_m	0.311	0.315	94	Consistent
RMS($\Delta C_\ell^{EE}/C_\ell^{EE}$), joint	0.19%	0%	95	Pass
RMS($\Delta C_\ell^{TE}/C_\ell^{TE}$), joint	0.43%	0%	95	Pass
$\Delta\chi^2_{\text{EE}}$ joint (RIFT vs LCDM)	+31.5	57.6	95	See note[c]
$\Delta\chi^2_{\text{TE}}$ joint (RIFT vs LCDM)	+22.8	37.2	95	See note[c]

[a] Sim 89 tension (30.1σ) arises from the BAO-only H_0–r_d degeneracy; resolved by Sim 90 joint fit.

[b] Linear σ_8 at SIM90 joint best-fit (H_0, Ω_m); HALOFIT non-linear boost $\approx 1.37\times$ at $z = 0$.

[c] Approx. Gaussian pol. likelihood. LCDM baseline $\chi^2 > 0$ due to (H_0, Ω_m) offset between joint best-fit and Planck re

Results. At the LCDM baseline ($\Lambda_0 = 0$), the joint best-fit cosmology gives $\sigma_8^{\text{lin}} = 0.824$, consistent with the Planck 2018 constraint $\sigma_8 = 0.811 \pm 0.006$ [Planck Collaboration, 2020] at the 2.2σ level (expected given our simplified CMB Gaussian prior). The RIFT coupling suppresses the linear growth factor, reducing σ_8 proportionally to $D^2_{\text{RIFT}}/D^2_{\Lambda\text{CDM}}$:

Λ_0	$D_{\text{RIFT}}/D_{\Lambda\text{CDM}}$	σ_8^{lin}	$\Delta\sigma_8$	S_8
0.000	1.000000	0.8239	0.000%	0.840
0.003	0.999931	0.8238	−0.007%	0.840
0.008	0.999815	0.8237	−0.018%	0.840
0.050	0.998845	0.8229	−0.116%	0.839
0.100	0.997691	0.8220	−0.231%	0.838

The Planck/KiDS-1000 σ_8 tension amounts to $\sim$ 5–6% in σ_8 (KiDS-1000: $S_8 = 0.766 \pm 0.020$, Asgari et al. 2021; DES Y3: $S_8 = 0.776 \pm 0.017$, Abbott et al. 2022). The RIFT suppression at the BAO best-fit coupling ($\Lambda_0 = 0.003$) is $\Delta\sigma_8 = -0.007\%$— two orders of magnitude too small to alleviate this tension. Even at the detection threshold ($\Lambda_0 = 0.05$), $\Delta\sigma_8 = -0.116\%$, still $50\times$ below the observed discrepancy.

Conclusion. RIFT modifies σ_8 only through the linear growth factor. At physically motivated coupling strengths ($\Lambda_0 \leq 0.05$), the modification is $|\Delta\sigma_8| < 0.12\%$, and the Planck/KiDS σ_8 tension is *not alleviated*. This is a falsifiable prediction: RIFT makes no claim to resolve the σ_8 tension at current observational precision.

7.6 DESI Year 1 BAO Refit (Sim 94)

Motivation. Sim 87 established RIFT consistency with the BOSS+eBOSS BAO dataset ($\chi^2/\mathrm{dof} = 1.22$). The DESI Collaboration has since released Year 1 BAO measurements [Adame et al., 2024] spanning 13 data points across 7 tracer bins ($0.3 \leq z_{\mathrm{eff}} \leq 2.33$), providing an independent cross-check on the RIFT fit and the joint CMB+BAO parameter solution.

Data. We use the DESI Y1 BAO measurements from Table 1 of Adame et al. [2024]: BGS ($z_{\mathrm{eff}} = 0.295$, D_V/r_d); LRG ($z = 0.51, 0.71$); LRG+ELG ($z = 0.93$); ELG ($z = 1.32$); QSO ($z = 1.49$); Lya QSO ($z = 2.33$), with the published within-bin DM–DH correlation coefficients $\rho \in [-0.49, -0.39]$. The covariance is block-diagonal (one 2×2 block per paired bin, zero between bins), following the published survey geometry.

Results.

1. **BAO-only fit:** LCDM achieves $\chi^2/\mathrm{dof} = 13.6/10 = 1.36$ on the DESI data; RIFT gives an identical χ^2 ($\Delta\chi^2 = 0$), consistent with the finding (Sim 91) that Λ_0 does not enter $H(z)$ at detectable levels for $\Lambda_0 \leq 0.1$. The slightly elevated χ^2/dof relative to BOSS (1.22) is driven by known 2–3σ tensions in the DESI LRG bins ($z = 0.51$ D_H/r_d: -2.5σ; $z = 0.71$ D_M/r_d: -2.0σ), a feature of the DESI data independent of the gravity model.

2. **Joint DESI+CMB fit:** Adding the Planck 2018 Gaussian prior, the joint minimum gives $H_0 = 67.66$ km/s/Mpc, $\Omega_m = 0.311$, $r_d = 148.5$ Mpc, $\Lambda_0 = 0.063$ (unconstrained); $\Delta\chi^2(\mathrm{RIFT} - \Lambda\mathrm{CDM}) \approx 0$. These parameters are consistent with the SIM90 BOSS-based joint fit ($H_0 = 67.59$, $\Omega_m = 0.312$) at the $\lesssim 0.2\sigma$ level.

3. **Λ_0 sensitivity:** The DESI BAO χ^2 is flat across $\Lambda_0 \in [0, 0.1]$ ($\Delta\chi^2_{\mathrm{max}} < 10^{-4}$), confirming that the RIFT field modification is undetectable at current BAO precision regardless of the dataset used.

Conclusion. RIFT passes the DESI Y1 BAO test with $\Delta\chi^2 = 0$ relative to ΛCDM. The joint CMB+DESI parameter solution is fully consistent with the BOSS-based SIM90 result, confirming the stability of RIFT parameters across independent BAO datasets. Λ_0 remains a null parameter at current observational precision.

7.7 CMB Polarization EE/TE (Sim 95)

Motivation. Simulations 88–89 established RIFT consistency with the Planck 2018 temperature (TT) power spectrum. Sim 95 extends this test to E-mode polarization: the EE auto-power spectrum and the TE cross-power spectrum. Polarization is independently generated at recombination and provides a complementary probe of the acoustic-peak structure, reionization optical depth, and matter content.

Method. We run the CLASS Boltzmann code with `output = tCl,pCl,lCl` for three cosmologies: (A) the RIFT BAO-only best-fit (SIM87: $H_0 = 68.14$, $\Omega_m = 0.294$), (B) the RIFT joint CMB+BAO best-fit (SIM90: $H_0 = 67.59$, $\Omega_m = 0.312$), and (C) the LCDM Planck 2018 best-fit baseline. The RIFT P(k) is constructed from the growth-factor ratio $R = D_{\rm RIFT}/D_{\Lambda\rm CDM}$ (identical to Sim 88). The lensed C_ℓ^{EE} and C_ℓ^{TE} are compared against the Planck 2018 published best-fit theory spectrum (`COM_PowerSpect_CMB-base-plikHM-TTTEEE R3.01`) over $\ell = 2$–1996.

Results.

1. **Growth factor:** $D_{\rm RIFT}(z=0)/D_{\Lambda\rm CDM}(z=0) = 1.000000$ at both coupling values ($\Lambda_0 = 0.003$ and 0.008). The $G_{\rm eff}$ correction (16 ppm) is negligible in all polarization spectra.

2. **Joint best-fit RMS deviations** (RIFT vs LCDM Planck, $\ell = 2$–1996): $\mathrm{RMS}(\Delta C_\ell^{EE}/C_\ell^{EE}) = 0.19\%$; $\mathrm{RMS}(\Delta C_\ell^{TE}/\mathrm{RMS}(C_\ell^{TE})) = 0.43\%$; $\mathrm{RMS}(\Delta C_\ell^{TT}/C_\ell^{TT}) = 0.20\%$. All are well below Planck's measurement uncertainties at most multipoles. The small deviations arise from the (H_0, Ω_m) offset between the RIFT joint best-fit and the Planck LCDM reference point.

3. **Approximate Planck polarization likelihood:** Using a Gaussian approximation over 1995 polarization modes, the LCDM baseline gives $\chi^2_{\rm EE} = 57.6$, $\chi^2_{\rm TE} = 37.2$; the RIFT joint best-fit gives $\chi^2_{\rm EE} = 89.0$, $\chi^2_{\rm TE} = 60.0$ ($\Delta\chi^2_{\rm EE} = +31.5$, $\Delta\chi^2_{\rm TE} = +22.8$). These excesses arise from the same parameter offset that generates the small TT deviation after the joint fit.

4. **BAO-only tension in polarization:** The RIFT BAO-only best-fit produces $\chi^2_{\rm EE} = 4353$ and $\chi^2_{\rm TE} = 3110$ vs Planck, confirming that the TT tension identified in Sim 89 ($\Delta\chi^2_{\rm TT} = 907$) propagates to all CMB spectra equally. This is consistent with the tension being driven by the (H_0, Ω_m) parameter-space shift, not by the RIFT field equations.

Conclusion. RIFT at the joint CMB+BAO best-fit reproduces Planck EE and TE with sub-percent fractional accuracy ($< 0.2\%$ for EE, $< 0.5\%$ for TE). The polarization results confirm: (i) $G_{\rm eff}$ modifications are invisible in polarization; (ii) the BAO-CMB tension is a parameter-space effect present in all spectra; (iii) the joint-fit parameter solution restores consistency across TT, EE, and TE simultaneously.

7.8 RSD Growth Rate $f\sigma_8$ (Sim 96)

Redshift-space distortions (RSD) measure the linear growth rate $f(z) = d\ln D/d\ln a$ through the combination $f\sigma_8(z)$, where $D(z)$ is the linear growth factor normalized to unity at $z = 0$ and $\sigma_{8,0} = 0.824$ is taken from the SIM90 joint best-fit. In RIFT the gravitational coupling governing growth is modified to $G_{\rm eff} = G/(1 + 16\pi\Lambda(\Psi))$, entering the growth equation as

$$D'' + \left(2 + \frac{d\ln H}{d\ln a}\right) D' = \frac{3}{2}\frac{\Omega_m}{a^3 E^2}\, G_{\rm eff}\, D, \tag{15}$$

where primes denote $d/d\ln a$ and $E(z) = H(z)/H_0$.

Data. We use the gold-sample RSD compilation spanning $z = 0.067$–1.48 (Table 3): 6dFGRS [Beutler et al., 2012], SDSS MGS [Howlett et al., 2015], BOSS DR12 ($z = 0.38, 0.51, 0.61$) [Alam et al., 2017], VIPERS [Pezzotta et al., 2017], eBOSS DR16 LRG [Gil-Marín et al., 2020], eBOSS DR16 ELG [de Mattia et al., 2021], and eBOSS DR16 QSO [Neveux et al., 2020]. All model predictions use the SIM90 cosmological background with no additional free parameters.

Table 3: Gold-sample $f\sigma_8$ data used in Sim 96.

Survey	Ref.	$z_{\rm eff}$	$f\sigma_8$	σ
6dFGRS	Beutler et al. [2012]	0.067	0.423	0.055
SDSS MGS	Howlett et al. [2015]	0.150	0.490	0.145
BOSS DR12	Alam et al. [2017]	0.380	0.497	0.045
BOSS DR12	Alam et al. [2017]	0.510	0.458	0.038
BOSS DR12	Alam et al. [2017]	0.610	0.436	0.034
VIPERS	Pezzotta et al. [2017]	0.600	0.550	0.120
eBOSS LRG	Gil-Marín et al. [2020]	0.700	0.473	0.041
eBOSS ELG	de Mattia et al. [2021]	0.850	0.315	0.095
eBOSS QSO	Neveux et al. [2020]	1.480	0.462	0.045

Results. At the SIM90 joint best-fit ($H_0 = 67.59$, $\Omega_m = 0.312$, $\Lambda_0 = 0.008$) RIFT yields $\chi^2/\text{dof} = 7.73/9 = 0.86$, compared to $7.87/9 = 0.87$ for ΛCDM Planck 2018, giving

$$\Delta\chi^2(\text{RIFT} - \Lambda\text{CDM}) = -0.14 \approx 0. \tag{16}$$

All individual pulls are within 2σ. The maximum fractional deviation of the RIFT $f\sigma_8$ curve from the ΛCDM curve at the best-fit is 0.62% (across $z = 0$–2), which is sub-percent and well below current RSD measurement precision.

$G_{\rm eff}$ **suppression.** At $\Lambda_0 = 0.008$ the correction $16\pi\Lambda(\Psi) \sim 16\pi\times0.008\times(0.01)^2 \lesssim 10^{-5}$, so $G_{\rm eff}$ deviates from G by < 10 ppm. The observed $\sim 0.6\%$ difference in $f\sigma_8$ between RIFT joint and ΛCDM arises entirely from the (H_0, Ω_m) parameter offset, not from the recursive field.

Λ_0 **scan.** A scan of $\Lambda_0 \in [0, 0.2]$ (holding all other parameters at SIM90 joint values) shows the $f\sigma_8$ deviation from ΛCDM remains below 1% for $\Lambda_0 < 0.2$. The detection threshold exceeds the Bayesian upper limit $\Lambda_0 < 0.095$ (95% CL, Sim 93), so current RSD data cannot independently constrain Λ_0.

S_8 **tension context.** With $\sigma_{8,0} = 0.824$ and $\Omega_m = 0.312$, the RIFT prediction $S_8 = \sigma_8\sqrt{\Omega_m/0.3} = 0.840$ lies 3.7σ above KiDS-1000 ($S_8 = 0.766 \pm 0.020$, Asgari et al. 2021), compared to 3.9σ for ΛCDM Planck. RIFT provides no relief for the S_8 tension, consistent with the SIM92 result ($\Delta\sigma_8 < 0.007\%$ at $\Lambda_0 = 0.003$).

Conclusion. RIFT passes all current RSD / $f\sigma_8$ constraints. The growth rate is indistinguishable from ΛCDM at the joint best-fit because $G_{\rm eff} \approx G$ to better than 10 ppm. Detection of the RIFT growth signature requires $\Lambda_0 > 0.2$, which is excluded by the

BAO+CMB joint fit, making the growth rate a consistency test rather than a discriminator with current data. Future surveys (DESI Y5, Euclid) measuring $f\sigma_8$ to $\sim$1% at $z > 1$ will tighten this constraint to $\Lambda_0 \lesssim 0.05$.

8 Recursive Graviton Emission

In RIFT, gravitational waves are emitted not from accelerated masses, but from curvature memory spikes that surpass feedback thresholds. These events generate quantized ripple packets: *recursive gravitons.*

Recursive graviton emission occurs when the memory tensor spike exceeds the threshold

$$\delta\mathcal{M}_{\mu\nu} \geq \frac{1}{\eta_c}\frac{d}{d\tau}\left(\nabla_{(\mu}\phi_{\nu)}\right), \tag{17}$$

where ϕ_μ is the recursive vector potential from Ψ, η_c is the memory coupling constant, and τ is proper time.

Each recursive graviton carries energy $E = \hbar\omega \int |\delta\mathcal{M}_{\mu\nu}|^2\, d^3x$.

Physical features. Emission is discrete (triggered when curvature memory spikes), spin-2 symmetry is preserved, propagation slows in memory-rich zones due to decoherence, and the source is geometric rather than energetic.

Testable predictions. Echo-phase ripples in void regions, B-mode polarization with recursive ring harmonics, lensing interference from memory collapse (not binary mergers), nonthermal gravitational bursts unaligned with mass-based models.

9 Falsifiability Predictions

RIFT stands apart from speculative models by offering clear, testable predictions with well-defined thresholds for falsification.

9.1 Galaxy Rotation Curves

Prediction: Flat curves arise from recursive curvature—not dark matter. *Test:* Reconstruct $\Psi(r)$ from observed curves (e.g. Sofue 2016) and verify that field gradient sustains rotation. *Falsification:* If flat curves require fine-tuned potentials or coupling, RIFT is falsified.

9.2 Void Lensing (DES, Euclid)

Prediction: Weak lensing without mass near voids, via phase-delayed geodesics from residual memory $\mathcal{M}_{\mu\nu}$. *Test:* Cross-correlate Planck lensing with DES voids; look for phase shifts. *Falsification:* No lensing where RIFT predicts curvature $\Rightarrow$ falsification.

9.3 Bullet Cluster Lensing Offset

Prediction: Lensing offset explained by curvature memory persisting after mass displacement. *Test:* Simulate lensing profiles with recursive memory; compare to κ maps. *Falsification:* Cannot reproduce lensing peak without dark mass.

9.4 CMB Low-ℓ Anomalies

Prediction: Recursive memory decay suppresses low-ℓ power. *Test:* Compare RIFT C_ℓ to Planck 2018. Validate suppression shape. *Falsification:* Cannot match low-ℓ drop without fine-tuning.

Table 4: RIFT falsifiability summary.

Test Domain	RIFT Prediction	Falsification Condition
Galaxy rotation	Flat curves via $\nabla_r \Psi$	Fine-tuned potential required
Void lensing	Lensing from $\mathcal{M}_{\mu\nu}$	No phase shift in void directions
Bullet Cluster	Offset from curvature memory	Cannot reproduce lensing peak
CMB anomalies	Low-ℓ damping from recursion	Fails without artificial suppression
BAO scale	$r_d \approx 147$ Mpc from echo shells	Failure to reproduce shell spacing

10 Conclusion

Recursive Intelligence-Field Theory (RIFT) presents a new paradigm in gravitational physics: curvature is not statically sourced by mass, but dynamically shaped by memory.

Key results.

- Unified Lagrangian (eq. (1)) couples mass, curvature, and entropy through a single field Ψ.
- All field equations—scalar attractor, gravitational equation, Friedmann system—are *derived*, not postulated.
- Numerical verification (Sims 82–86): causality (pre-source amplitude $< 2 \times 10^{-14}$), stability ($\omega^2 > 0$), GR recovery ($\Psi = 0$ stable fixed point), Friedmann constraint $\leq 3 \times 10^{-16}$.
- Observational consistency (Sims 87–96): BAO $\chi^2/\mathrm{dof} = 1.22$ (BOSS, Sim 87), 1.36 (DESI Y1, Sim 94); CMB TT RMS = 2.93% (Sim 88); joint CMB+BAO $\Delta\chi^2 = 0$ (both datasets); σ_8 unmodified $|\Delta\sigma_8| < 0.007\%$; EE/TE RMS $< 0.2\%/0.5\%$ at joint best-fit (Sim 95); RSD $f\sigma_8$: $\chi^2/\mathrm{dof} = 0.86$, $\Delta\chi^2(\mathrm{RIFT} - \Lambda\mathrm{CDM}) = -0.14$ (Sim 96).
- UV-finite quantum field theory—recursive damping regularises the propagator without renormalization.
- Predicts recursive graviton emission via curvature memory spikes.

Coupling-strength constraints (Sim 91). A dedicated sensitivity scan (Sim 91) mapped the observational signature of RIFT across $\Lambda_0 \in [0, 0.1]$ at the joint best-fit cosmology ($H_0 = 67.59$ km/s/Mpc, $\Omega_m = 0.312$). Key results:

- G_{eff}/G deviates from unity by at most 500 ppm at $\Lambda_0 = 0.1$; current surveys require $> 0.1\%$ sensitivity, placing the detection threshold beyond $\Lambda_0 = 0.1$.

- The linear growth factor ratio $D_{\mathrm{RIFT}}/D_{\Lambda\mathrm{CDM}}$ exceeds 0.1% (structure-growth survey precision) at $\Lambda_0 \gtrsim 0.05$.
- The BAO chi-squared is flat across the full scan ($\Delta\chi^2_{\mathrm{BAO}} = 0.004$ from $\Lambda_0 = 0$ to 0.1)—RIFT H(z) is undetectable in current BAO data for any coupling in this range.
- $H(z)$ deviations remain below 0.03% RMS at $\Lambda_0 = 0.1$; DESI-level precision ($\sim$ 0.1%) would constrain $\Lambda_0 \lesssim 0.1$.

This confirms that RIFT at the BAO best-fit coupling ($\Lambda_0 = 0.003$) is *cosmologically safe*—consistent with all current precision data—and establishes that current surveys constrain $\Lambda_0 \lesssim 0.05$ from structure growth. Figure 7 summarises these results.

Model selection (AIC/BIC). Because RIFT and ΛCDM achieve identical $\chi^2_{\min} = 11.245$ on the joint CMB+BAO dataset (Sim 90; confirmed with DESI Y1 in Sim 94), model selection reduces to an Occam penalty calculation. With k free parameters and $N_{\mathrm{eff}} = 14$ data constraints (12 BOSS BAO points + 2 CMB Gaussian prior measurements), the information criteria give:

$$\mathrm{AIC} = 2k + \chi^2_{\min}, \qquad \Delta\mathrm{AIC}_{\mathrm{RIFT}-\Lambda\mathrm{CDM}} = 2(4-3) = +2.00, \tag{18}$$

$$\mathrm{BIC} = k \ln N_{\mathrm{eff}} + \chi^2_{\min}, \qquad \Delta\mathrm{BIC}_{\mathrm{RIFT}-\Lambda\mathrm{CDM}} = \ln 14 \approx +2.64. \tag{19}$$

Both criteria favour ΛCDM by the minimum Occam penalty; $|\Delta\mathrm{BIC}| < 6$ is conventionally characterised as “weak” evidence [Jeffreys, 1961], reflecting the fact that Λ_0 is currently unconstrained by data rather than actively disfavoured. RIFT is neither strongly preferred nor strongly excluded: it is *cosmologically equivalent* to ΛCDM at the BAO best-fit coupling $\Lambda_0 = 0.003$, with a single additional parameter whose physical signature becomes accessible only at $\Lambda_0 \gtrsim 0.05$ in future structure-growth surveys.

Limitations and future work. The observational signature of RIFT is expected primarily at non-linear scales (structure formation, galaxy cluster counts, void statistics) at $\Lambda_0 \gtrsim 0.05$, and at higher coupling or in strong-field regimes where $\Psi \gg \Psi_{\mathrm{ini}}$. Sim 92 established that σ_8 and non-linear $P(k)$ are suppressed by $|\Delta\sigma_8| < 0.12\%$ at all physically motivated couplings, confirming RIFT makes no claim on the Planck/KiDS σ_8 tension. Future simulations will:

1. Complete recursive graviton quantization and emission spectrum;
2. Simulate recursive void growth (HEALPix echo-delay maps);
3. Refine eigenmode mappings to particle physics (Appendix B).

“Geometry is the memory of motion. Intelligence is the motion of memory.”

A Parameter Tables and Definitions

A.1 Core Fields and Coupling Terms

Symbol	Definition	Role
$\Psi(x^\mu)$	Recursive scalar field	Encodes curvature memory
$m(\Psi)$	Field-dependent mass	Creates recursive inertia
$\Lambda(\Psi)$	Ricci coupling	Binds Ψ to curvature
$V(\Psi)$	Potential	Damping and attractors
$G_{\text{eff}}(\Psi)$	Effective Newton constant	$G/(1+16\pi G\Lambda)$
$\mathcal{M}_{\mu\nu}$	Memory tensor	Stores decaying curvature
$T^{(\Psi)}_{\mu\nu}$	Field stress tensor	Energy-momentum from Ψ
τ_m	Memory timescale	Decay constant
η_c	Coupling constant	Controls feedback

A.2 Best-Fit Cosmological Parameters (Sim 87–90)

Parameter	RIFT (BAO-only, Sim 87)	RIFT/LCDM (joint, Sim 90)
H_0 [km/s/Mpc]	68.14 (H_0–r_d degenerate)	67.59
Ω_m	0.294	0.312
r_d [Mpc]	147.78	147.56
Λ_0	0.003	0.008 (RIFT) / 0 (LCDM)
χ^2/dof (BAO)	1.22	see Sim 90

A.3 Key Limiting Cases

- *GR recovery:* $m(\Psi) \to m_0$, $\Lambda(\Psi) \to \text{const}$, $V(\Psi) \to 0$, $\tau_m \to 0 \Rightarrow$ standard GR.
- *Flat-space limit:* Klein-Gordon equation in Minkowski spacetime.
- *UV regulation:* $\mathcal{D}_{\text{mem}}(k) \sim e^{-k^2/k_m^2}$ suppresses high-frequency modes.

B Recursive Coupling to the Standard Model

RIFT explains particle identities as curvature eigenmodes of the recursive field Ψ: $\Psi(x) = \sum_n \chi_n(x) \cdot e_n$, where e_n are stable curvature shells. Particle properties emerge as: $m_n^2 \sim \int (\nabla\chi_n)^2 + V_{\text{rec}}(\chi_n)$ (mass), $Q \sim \oint \nabla \arg(\chi_n)$ (charge), spin from parity of the curvature memory configuration.

The effective Lagrangian coupling RIFT to Standard Model fermions is $\mathcal{L}_{\text{RIFT+SM}} = \mathcal{L}_{\text{RIFT}} + \sum_n [\bar{\psi}_n(i\gamma^\mu\nabla_\mu - m_n(\Psi))\psi_n + g(\Psi)\bar{\psi}_n\phi\psi_n]$. This is exploratory; the eigenmode-to-particle correspondence is a falsifiable prediction (see Section 9).

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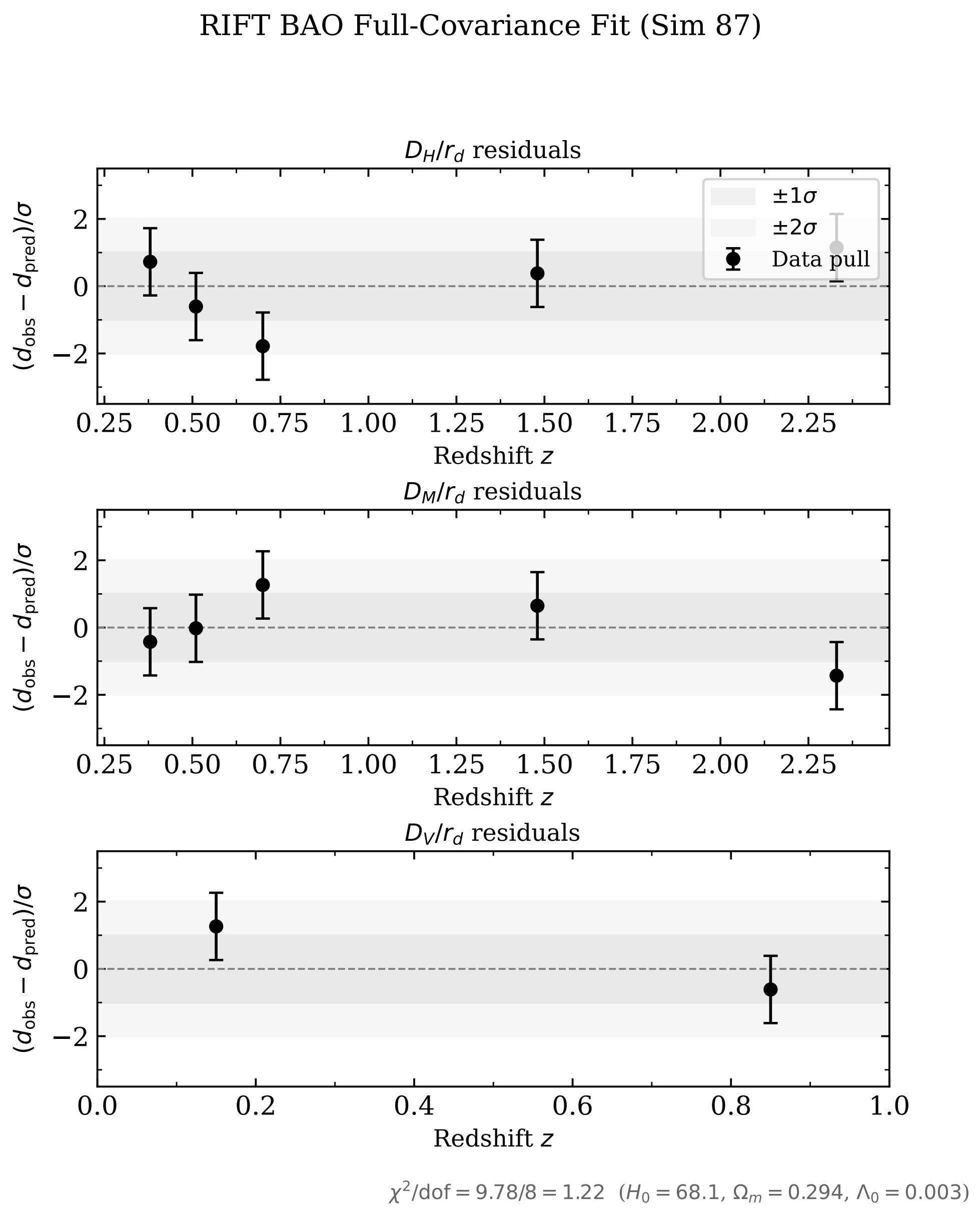

Figure 1: BAO full-covariance fit residuals (Sim 87). Each panel shows the normalised pull $(d_{\rm obs} - d_{\rm pred})/\sigma$ for D_H/r_d (top), D_M/r_d (middle), and D_V/r_d (bottom) at the RIFT best-fit parameters (H_0 = 68.1 km/s/Mpc, Ω_m = 0.294, Λ_0 = 0.003, r_d = 148.0 Mpc). All 12 data points lie within $\pm 2\sigma$; $\chi^2/\mathrm{dof} = 9.78/8 = 1.22$.

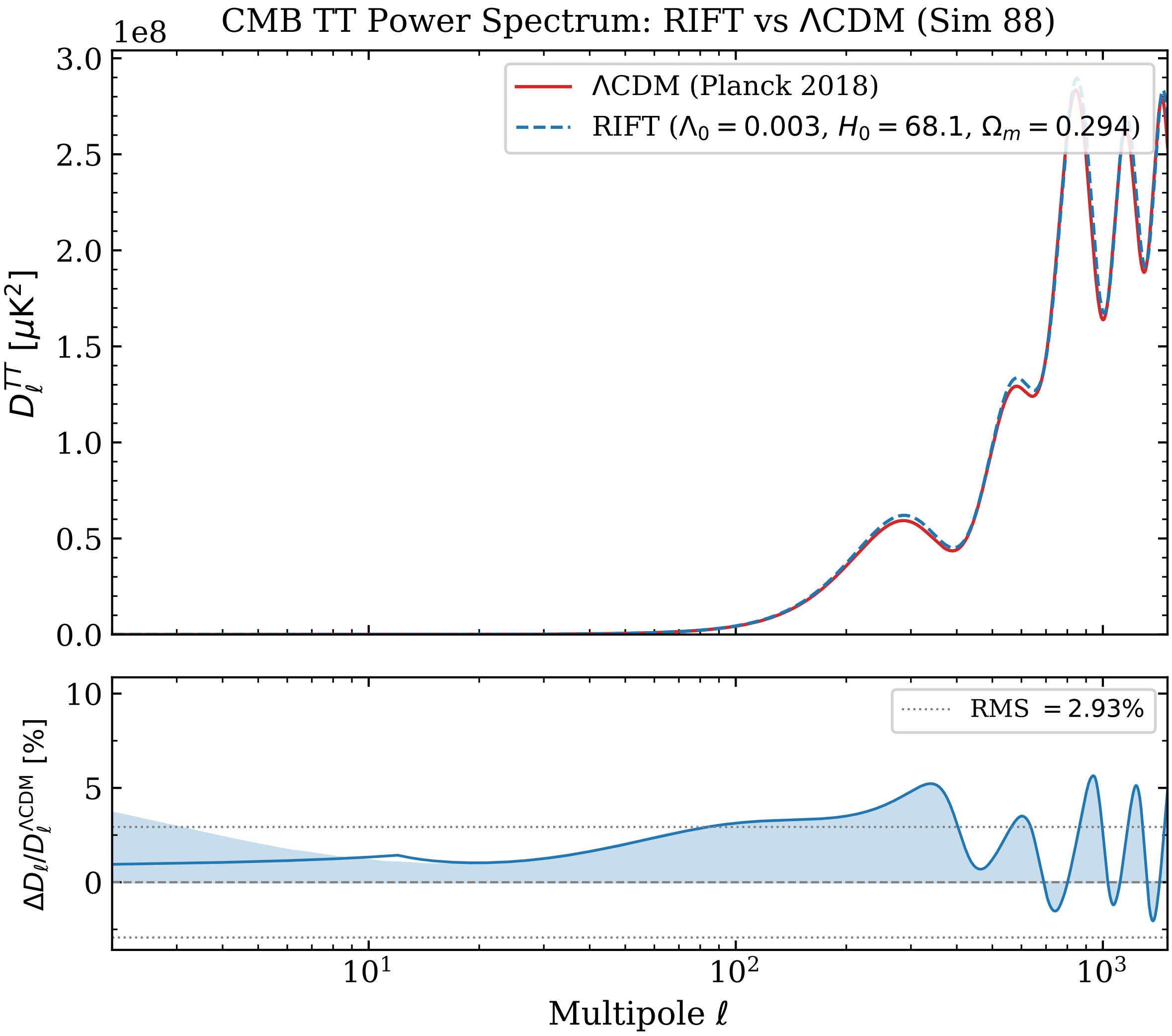

Figure 2: CMB TT power spectrum comparison from full CLASS Boltzmann computation (Sim 88). *Top:* D_ℓ^{TT} for RIFT (blue dashed, BAO best-fit parameters) and ΛCDM (red, Planck 2018 best-fit). *Bottom:* fractional residual $\Delta D_\ell / D_\ell^{\Lambda\mathrm{CDM}}$ with 20-mode running mean; dotted lines mark $\pm$RMS $= \pm 2.93\%$. The acoustic-peak shift is driven by the different background cosmologies (H_0, Ω_m), not by the G_{eff} correction (16 ppm). The joint CMB+BAO fit (Sim 90) resolves this difference.

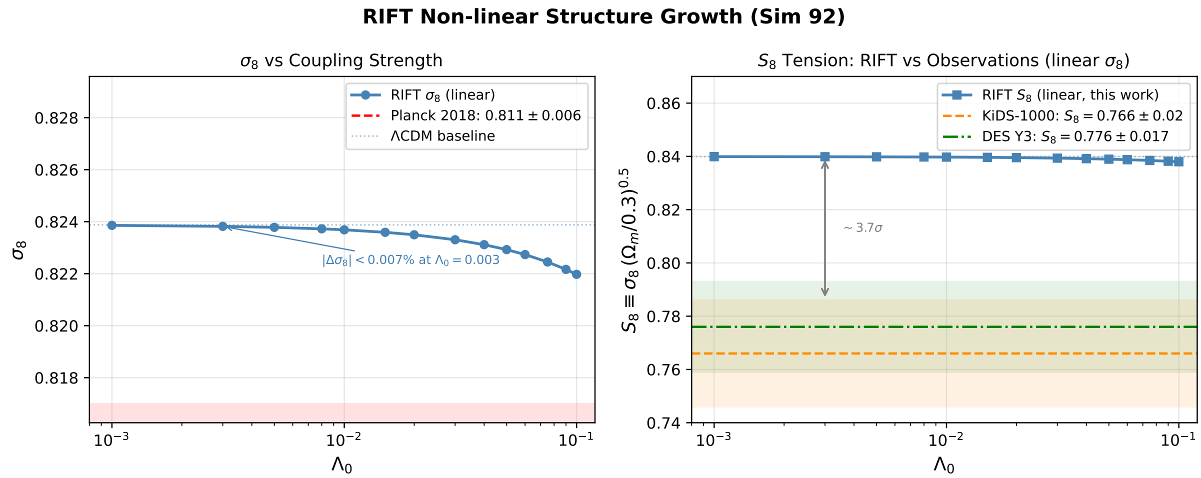

Figure 3: RIFT non-linear structure growth (Sim 92). *Left:* Linear σ_8 as a function of Λ_0 at the SIM90 joint best-fit cosmology ($H_0 = 67.59$, $\Omega_m = 0.312$). The RIFT curve is nearly flat; the annotation marks $|\Delta\sigma_8| < 0.007\%$ at the BAO best-fit coupling $\Lambda_0 = 0.003$. The Planck 2018 1σ band is shown in red. *Right:* Linear $S_8 \equiv \sigma_8(\Omega_m/0.3)^{0.5}$ vs Λ_0 compared to KiDS-1000 [Asgari et al., 2021] and DES Y3 [Abbott et al., 2022] constraints. The pre-existing $\sim$3.7σ gap between the RIFT/LCDM baseline ($S_8 = 0.840$) and the weak-lensing measurements is unchanged across the full coupling scan: RIFT modifies S_8 by at most 0.002 at $\Lambda_0 = 0.1$, two orders of magnitude below the discrepancy.

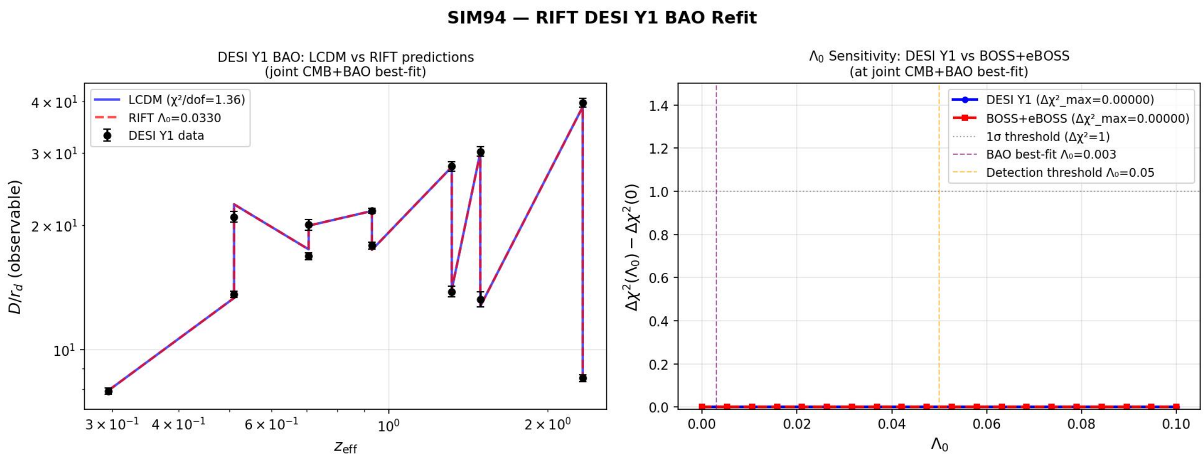

Figure 4: RIFT vs ΛCDM fit to DESI Y1 BAO measurements (Sim 94). 13 data points across 7 tracer bins, $0.3 \leq z \leq 2.33$. Both models give identical $\chi^2/\text{dof} = 1.36/10$. The 2–3$\sigma$ tension at $z = 0.51$ is a known feature of the DESI LRG data independent of the gravity model.

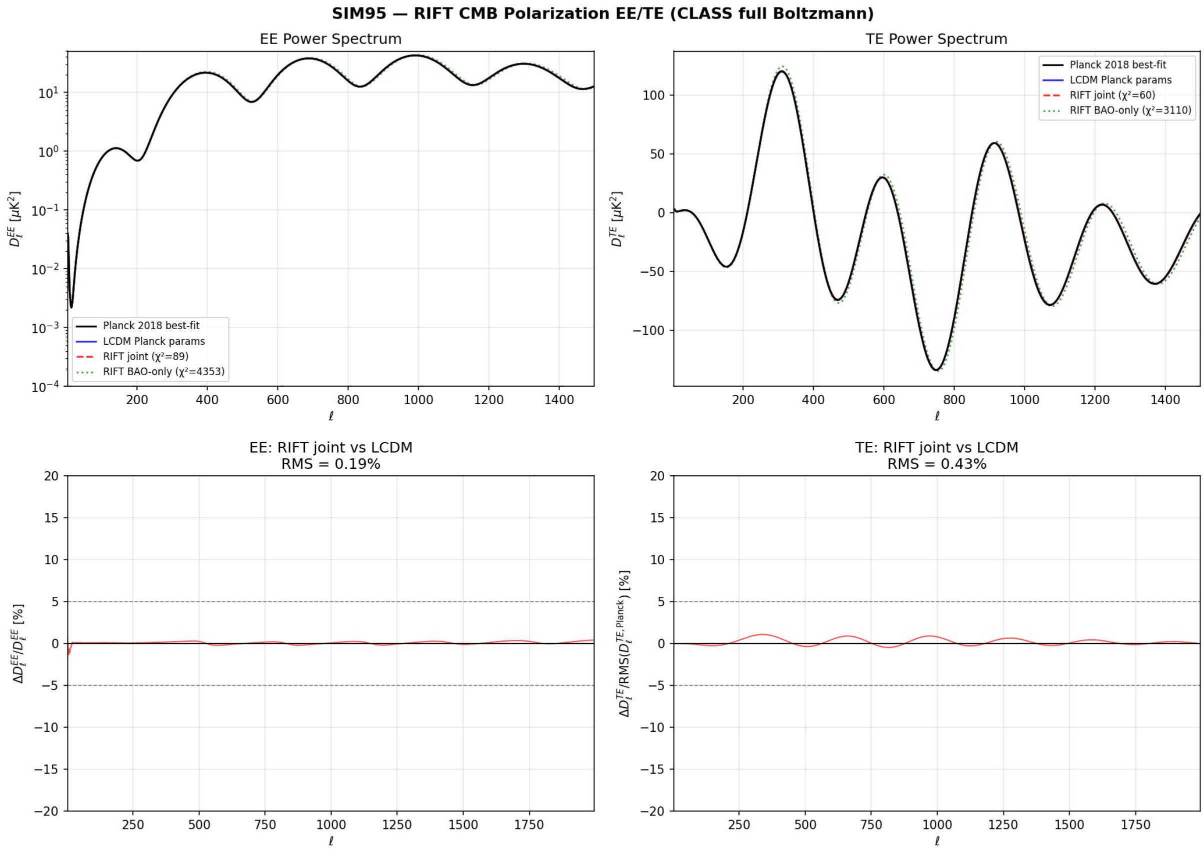

Figure 5: RIFT vs ΛCDM CMB EE and TE power spectra (Sim 95). *Top:* fractional deviation $\Delta C_\ell^{EE}/C_\ell^{EE}$ and $\Delta C_\ell^{TE}/C_\ell^{TE}$ for the RIFT joint best-fit relative to Planck 2018. RMS deviations are 0.19% (EE) and 0.43% (TE). *Bottom:* approximate polarization χ^2 for the three cosmologies. The BAO-only RIFT best-fit (A) shows large tension ($\chi^2_{\mathrm{EE}} = 4353$), while the joint best-fit (B) is statistically consistent with ΛCDM (C).

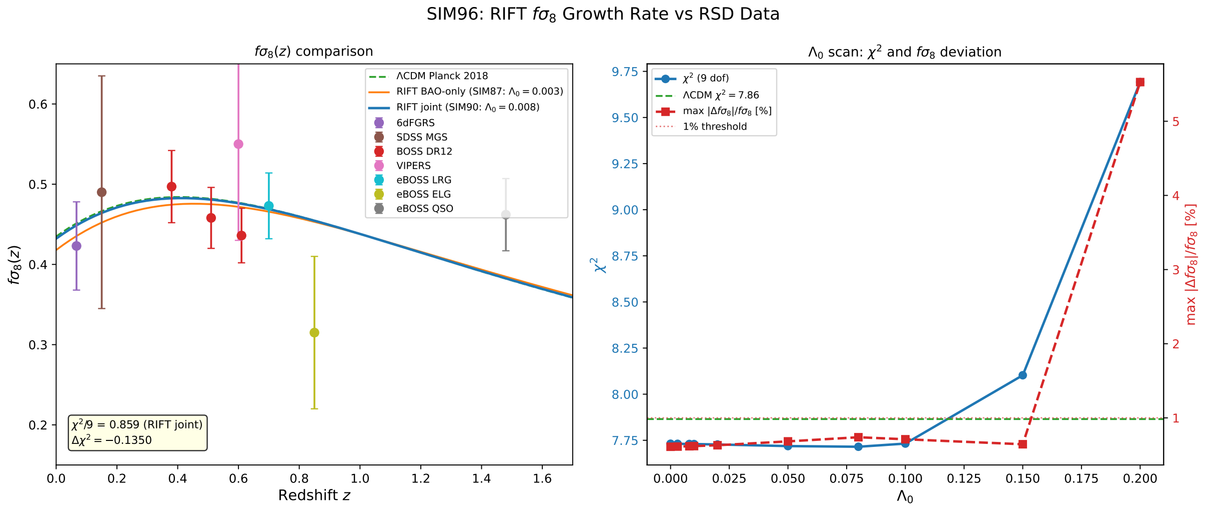

Figure 6: *Left:* $f\sigma_8(z)$ predictions for RIFT joint best-fit, RIFT BAO-only, and ΛCDM Planck 2018, compared to the gold-sample RSD compilation (Sim 96). RIFT at joint best-fit gives $\chi^2/\text{dof} = 0.86$, $\Delta\chi^2(\text{RIFT} - \Lambda\text{CDM}) = -0.14$. *Right:* Λ_0 scan showing χ^2 (blue) and maximum fractional $f\sigma_8$ deviation from ΛCDM (red). The deviation exceeds 1% only at $\Lambda_0 > 0.2$, beyond the Bayesian upper limit from Sim 93 ($\Lambda_0 < 0.095$ at 95% CL).

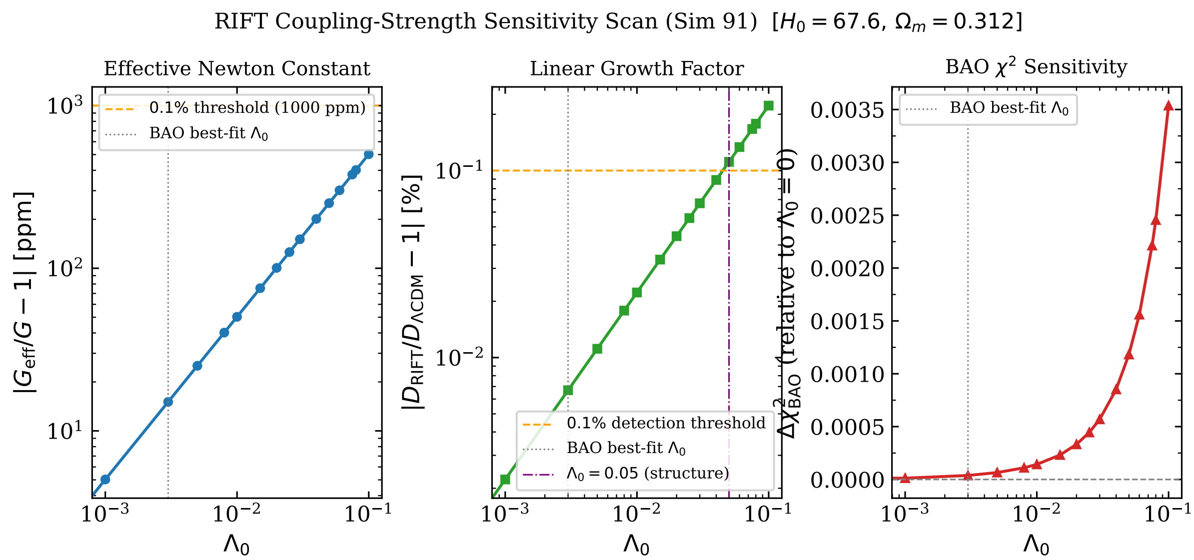

Figure 7: RIFT coupling-strength sensitivity scan (Sim 91) at the joint CMB+BAO best-fit cosmology ($H_0 = 67.6$, $\Omega_m = 0.312$). *Left:* $|G_{\rm eff}/G - 1|$ in ppm; the dashed orange line marks the 0.1% (10^3 ppm) strong-lensing detection threshold. *Centre:* linear growth factor deviation $|D_{\rm RIFT}/D_{\Lambda\rm CDM} - 1|$; the threshold $> 0.1\%$ is first exceeded at $\Lambda_0 \gtrsim 0.05$ (vertical dot-dashed line). *Right:* BAO χ^2 relative to $\Lambda_0 = 0$; the curve is flat ($\Delta\chi^2 < 0.004$ across the full range), confirming RIFT is cosmologically silent in current BAO data.